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Rewarding the Graph Behind the Chain: Topology-Aware Process Supervision for RL and Reasoning Distillation

 (/pdf?id=2JqC3rt4AV)

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 23 May 2026 (modified: 03 Jul 2026)  ACL ARR 2026 May Submission
 May, Senior Area Chairs, Area Chairs, Reviewers, Authors, Secondary Reviewers
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Keywords: process reward models, reinforcement learning, knowledge distillation, mathematical reasoning, chain-of-thought

Abstract:

Long chain-of-thought reasoning in LLMs is neither strictly correct-or-wrong nor purely linear. Outcome-only rewards overlook unjustified intermediate steps, while existing process reward models depend on step-level annotations and fail to capture the implicit conditional dependencies within a reasoning chain. We propose TopoPRM, a topology-aware implicit process reward model that extracts a dependency DAG from each reasoning trace and integrates conditional dependency and sequential continuity as complementary components of a multi-source reward signal. Building on topoprpm, we design TGSD, a post-training framework that combines supervised warm-start, GRPO optimization under a correctness-first hierarchical reward, and topology-guided on-policy distillation. Across multiple public benchmarks, our method improves average accuracy over standard GRPO at both 7B and 9B scales while generating up to 2% fewer tokens, with consistent gains on elementary and general-domain reasoning.

Paper Type: Long

Research Area: NLP and Symbolic Reasoning

Research Area Keywords: Machine Learning for NLP, NLP and Symbolic Reasoning

Contribution Types: Model analysis & interpretability, Approaches to low-compute settings (efficiency), Publicly available software and/or pre-trained models

Languages Studied: English

Reassignment Request Area Chair: This is not a resubmission

Reassignment Request Reviewers: This is not a resubmission

A1 Limitations Section: This paper has a limitations section.

A2 Potential Risks: Yes

A2 Elaboration: This paper has a limitations section.

B Use Or Create Scientific Artifacts: Yes

B1 Cite Creators Of Artifacts: Yes

B1 Elaboration: §4 Experimental Part

B2 Discuss The License For Artifacts: No

B2 Elaboration: We use publicly available models under their original licenses but do not explicitly discuss license terms in the paper.

B3 Artifact Use Consistent With Intended Use: Yes

B3 Elaboration: §4.1
B4 Data Contains Personally Identifying Info Or Offensive Content: N/A
B5 Documentation Of Artifacts: Yes
B5 Elaboration: Appendix A.
B6 Statistics For Data: Yes
B6 Elaboration: §4.1 and Appendix A.
C Computational Experiments: Yes
C1 Model Size And Budget: Yes
C1 Elaboration: Appendix A
C2 Experimental Setup And Hyperparameters: Yes
C2 Elaboration: Appendix A
C3 Descriptive Statistics: Yes
C3 Elaboration: We report pass@1, pass@5, maj@5, Acc/kToken, and so on.
C4 Parameters For Packages: Yes
C4 Elaboration: §4.1 Setup and Appendix A
D Human Subjects Including Annotators: No
D1 Instructions Given To Participants: N/A
D2 Recruitment And Payment: N/A
D3 Data Consent: N/A
D4 Ethics Review Board Approval: N/A
E Ai Assistants In Research Or Writing: No
E1 Information About Use Of Ai Assistants: N/A
Author Submission Checklist: yes
EMNLP 2026 AI Reviewing Experiment: no
TLDR: 👁 We extract dependency graphs from reasoning traces and use their topology as a free process reward signal for both RL optimization and distillation.
Preprint: 👁 no
Preprint Status: 👁 We are considering releasing a non-anonymous preprint in the next two months (i.e., during the reviewing process).
Preferred Venue: 👁 EMNLP
Visa Needs: 👁 yes
Consent To Share Data: 👁 yes
Consent To Share Submission Details: 👁 On behalf of all authors, we agree to the terms above to share our submission details.
Association For Computational Linguistics - Blind Submission License Agreement: 👁 On behalf of all authors, I agree
Submission Number: 4012

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Official Review of Submission4012 by Reviewer HxUk

Official Review by Reviewer HxUk 📅 05 Jul 2026, 00:14 (modified: 08 Jul 2026, 20:14)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer HxUk

📄 Revisions (/revisions?id=Qj0r8hDJNm)

Paper Summary:

This paper proposes TopoPRM, which models long reasoning traces as implicit dependency DAGs and combines global topology with local sequential continuity for process rewards. Based on this reward model, TGSD integrates supervised warm-start, GRPO, and topology-guided on-policy distillation, improving overall accuracy while reducing reasoning length across several benchmarks.

Summary Of Strengths:

- 1. Modeling non-local dependencies between reasoning steps offers a meaningful alternative to conventional PRMs that treat reasoning as a purely linear sequence.
- 2. The correctness-first hierarchical reward and ACE mechanism help prevent structurally coherent but incorrect solutions from receiving overly high rewards. The motivation is clear and supported by the ablation results.
- 3. The experiments cover diverse reasoning tasks and examine accuracy, token efficiency, training dynamics, component contributions, and cross-scale distillation. Overall, the results suggest improvements in both performance and reasoning efficiency.

Summary Of Weaknesses:

- 1. Since the dependency edges are identified through expression reuse, variable overlap, and LLM-based inference, it would be useful to compare them with human-annotated dependency graphs or report edge-level precision, recall, and agreement. This would make it clearer whether the topology reward reflects the actual logical dependencies in the reasoning traces.
- 2. The distillation experiments mainly compare TGSD with standard SFT distillation and off-policy KL regularization. Comparisons with stronger on-policy distillation, reasoning compression, or explicit length-control methods under comparable data and training budgets could provide a more convincing assessment.
- 3. Most experiments are conducted within the Qwen model family and its distilled variants. Evaluation on other model families, such as Llama, could better support the generality of the method. In addition, some competition benchmarks are relatively small, but results over multiple random seeds or significance tests are not reported.

Comments Suggestions And Typos:

N/A

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3.5

Excitement: 3.5

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Limitations And Societal Impact:

yes

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 1 = No usable datasets submitted.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

Official Review of Submission4012 by Reviewer B5w7

Official Review by Reviewer B5w7 📅 03 Jul 2026, 19:35 (modified: 08 Jul 2026, 20:14)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer B5w7

📄 Revisions (/revisions?id=jPJSliyEbs)

Paper Summary:

This paper addresses process supervision for long chain-of-thought reasoning. It argues that outcome-only rewards and conventional step-level process reward models miss the non-linear dependency structure among reasoning steps. The proposed TopoPRM extracts an implicit dependency DAG from each reasoning trace and scores both global topology quality and local continuity. The authors then build TGSD, a post-training framework combining supervised warm-start, GRPO with a correctness-first hierarchical reward, and topology-guided on-policy distillation. Experiments on several public reasoning benchmarks at 7B and 9B scales report improvements over standard GRPO while also reducing response length.

Summary Of Strengths:

1. The paper studies an interesting direction: treating long CoT as a structure with non-linear dependencies rather than a purely linear step sequence. This is relevant to process supervision and RLVR.
2. TopoPRM converts reasoning traces into dependency DAGs and combines topology quality, continuity, final correctness, and length regularization in a GRPO-based training framework.
3. The experiments cover several public mathematical and general reasoning benchmarks, and report both accuracy and token efficiency. The results suggest average gains over outcome-only GRPO in several settings.

Summary Of Weaknesses:

1. The main uncertainty is the quality of the extracted dependency DAGs. TopoPRM relies on automatically extracting graph structure from free-form reasoning traces, but the paper does not fully establish that these edges correspond to true logical dependencies. If the graph is driven by expression reuse, references, or surface continuity, it may reward formally coherent but semantically wrong reasoning.
2. The relation between topology reward and semantic correctness remains unclear. The authors acknowledge that outcome-only GRPO can outperform TopoPRM on some competition-level benchmarks at 9B. This suggests that structurally clean traces do not necessarily imply stronger key deductions. The current results may show better trace regularity and token efficiency more than reliable process supervision.
3. The full system combines SFT warm-start, hierarchical-reward GRPO, ACE, topology extraction, and on-policy distillation. Although ablations are provided, it is still hard to isolate whether the main gains come from topology-aware supervision, length control, continuity regularization, the training recipe, or distillation. Comparisons with stronger length-aware GRPO, PRM, or verifier-guided baselines would make the claim more convincing.

Comments Suggestions And Typos:

1. A small human-labeled or otherwise verifiable evaluation of extracted dependency edges would strengthen the DAG extraction claim.
2. Error cases with high topology scores but wrong answers, and correct answers with low topology scores, would clarify the limits of the reward.
3. More compute-matched comparisons with length-aware RL, process verifiers, and outcome-level reranking baselines would improve the empirical evaluation.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Limitations And Societal Impact:

The paper already notes that topology is not a proof-level verifier. The limitations would be stronger with a quantitative analysis of the structure-semantic gap and a clearer statement of where the method is expected to work or fail.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 1 = No usable datasets submitted.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

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Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

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Official Review of Submission4012 by Reviewer TsKG

Official Review by Reviewer TsKG  02 Jul 2026, 23:12 (modified: 08 Jul 2026, 20:14)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer TsKG

 Revisions (/revisions?id=FyO6DjdUo)

Paper Summary:

The paper extracts a dependency graph from a reasoning trace using a rule-plus-LLM extractor, scores its topology and continuity, and folds those scores multiplicatively into a reward that also includes outcome, format and length terms. Around this it builds a three-stage pipeline: an SFT warm start, GRPO with an asymmetric clipping advantage estimator, and a topology-guided self-distillation step. It reports pass@5 gains over GRPO on nine benchmarks, a 15 to 24 percent token reduction, and cross-scale distillation.

Summary Of Strengths:

1. The premise is interesting: long chains of thought have non-linear dependency structure that step-local PRMs miss, and reusing one graph scorer for both reward shaping and distillation is an appealing unification.
2. The diagnostic suite is broad, covering token efficiency, collapse rate, reward-aggregation comparisons and cross-scale distillation.
3. The correctness-first multiplicative aggregation and the stratified advantage estimator are a sensible response to reward hacking in additive multi-signal rewards.

Summary Of Weaknesses:

1. The extractor at the center of the method is never validated against ground truth. All of the structural-quality gains are self-referential, and the paper itself admits the diagnostics are computed by the same extractor that drove training, so they are internal-consistency checks rather than an independent metric. There is no precision or recall of the extracted edges against human or verifier-derived support. The whole method rests on an unvalidated heuristic.
2. There are internal inconsistencies serious enough to undermine the empirical claims. The reported reasoning length is described as staying under 500 tokens in the text while the figure and a table give roughly 1568 and 1692. The main table mixes quoted and reproduced numbers and contains implausible rows, for instance an R1-distilled base scoring in the thirties on MATH and then jumping into the nineties after SFT, when that base should already be around ninety, which inflates the reported deltas. Some model names are non-standard, and one row sits far above its neighbors with no explanation.
3. The ablations look like brittleness rather than complementarity. Removing the continuity term drops the average from the mid-forties to the mid-teens, below outcome-only GRPO, with the collapse rate rising sharply. A reward whose ablation leaves the model far worse than using no process reward at all points to mis-specification or hackability, not to complementary signals.

Comments Suggestions And Typos:

- Separate the two settings clearly and state exactly what the outcome reward is on each public math benchmark. If the rubric reward was used there, the results need re-running with an answer-correctness reward.
- Explain why removing continuity drops below outcome-only GRPO. Is the topology term being hacked in that condition? A direct measurement of high-structure wrong-answer traces before and after the estimator would help.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.

Datasets: 2 = Documentary: The new datasets will be useful to study or replicate the reported research, although for other purposes they may have limited interest or limited usability. (Still a positive rating)

Software: 2 = Documentary: The new software will be useful to study or replicate the reported research, although for other purposes it may have limited interest or limited usability. (Still a positive rating)

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

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